

Week 1:

Project Data Preprocessing & Exploratory Data Analysis

***AnalystLabAfrica
Machine Learning Internship***

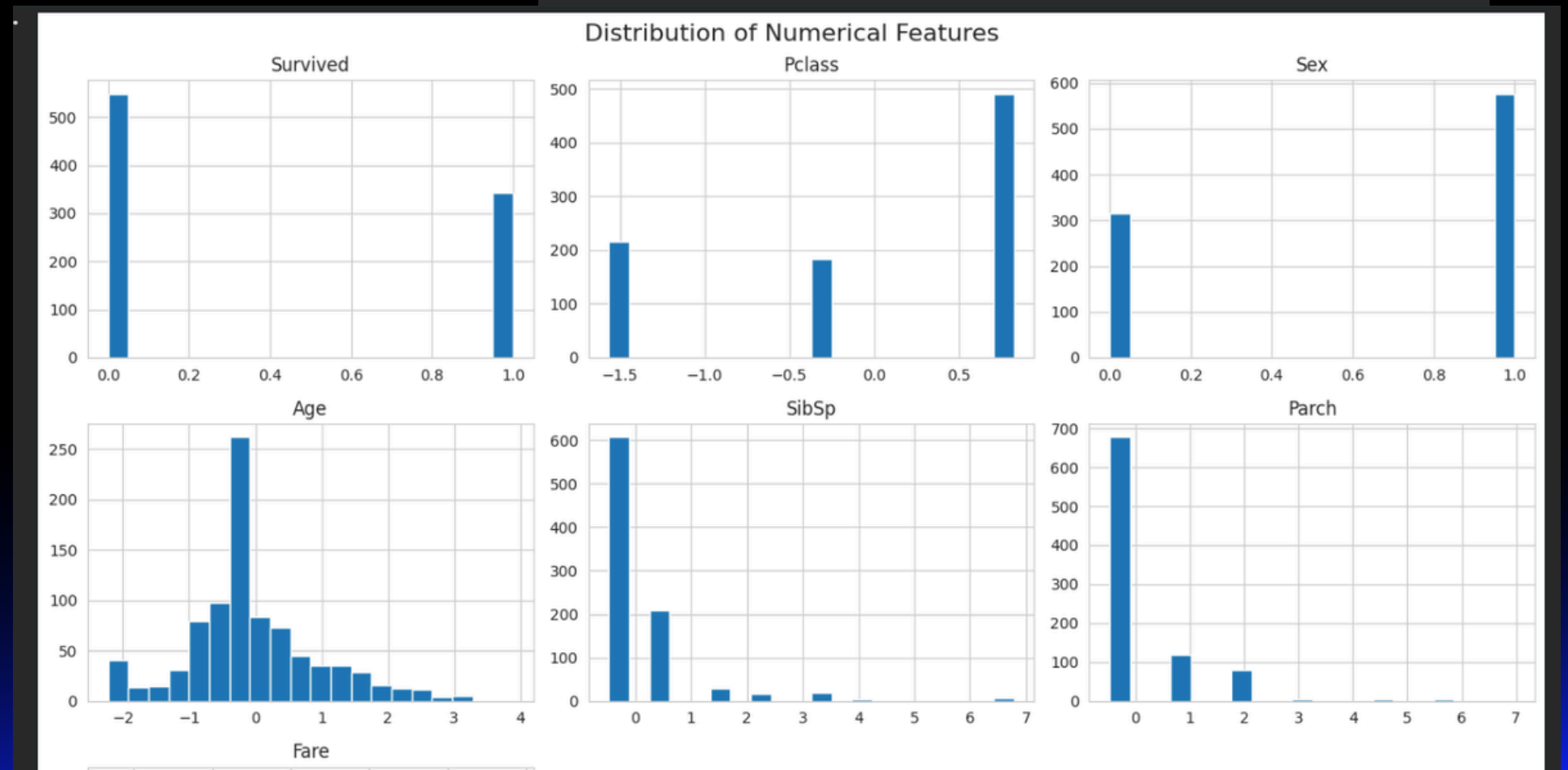
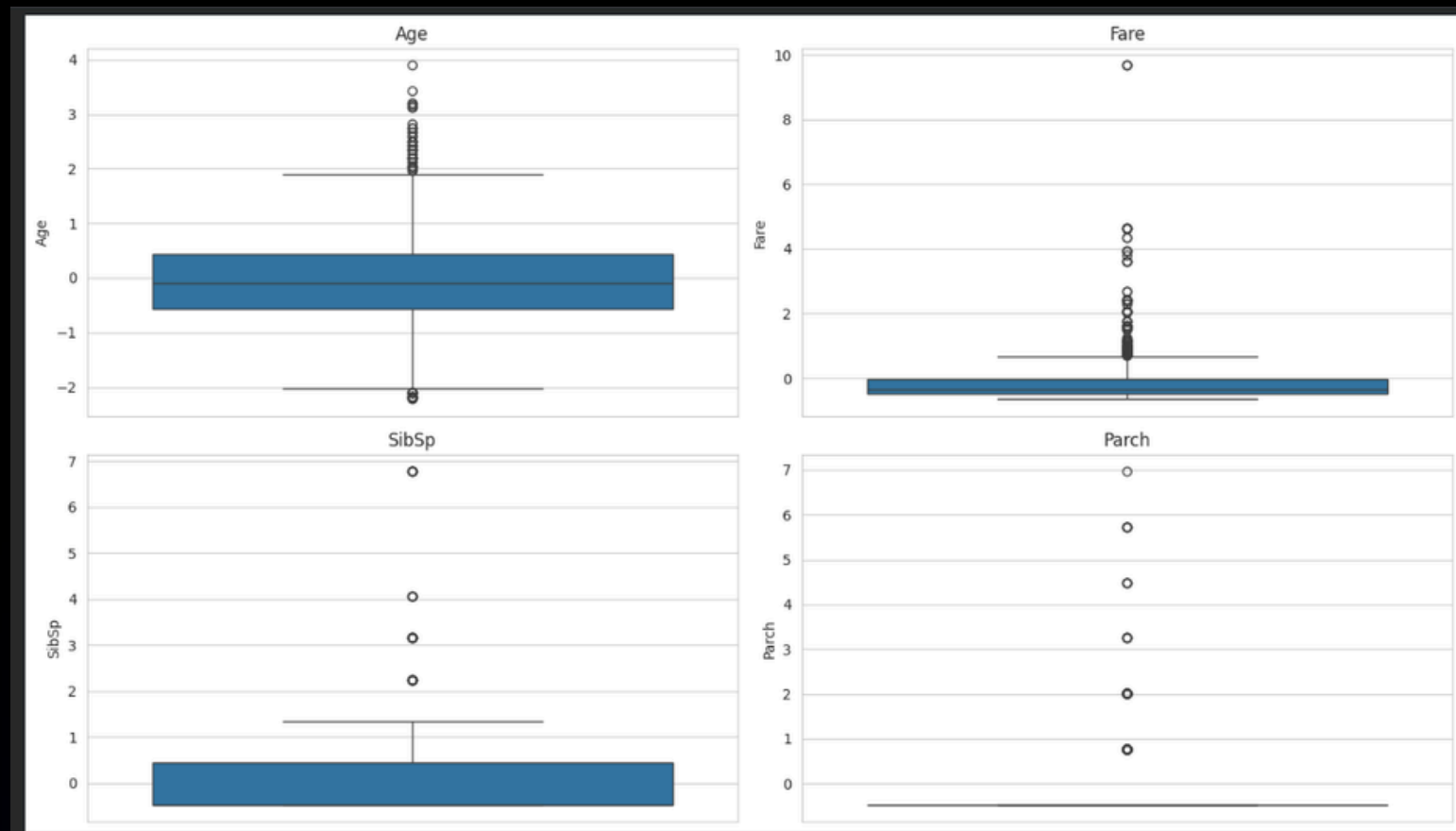
MARYAM ARIF

Data Preprocessing

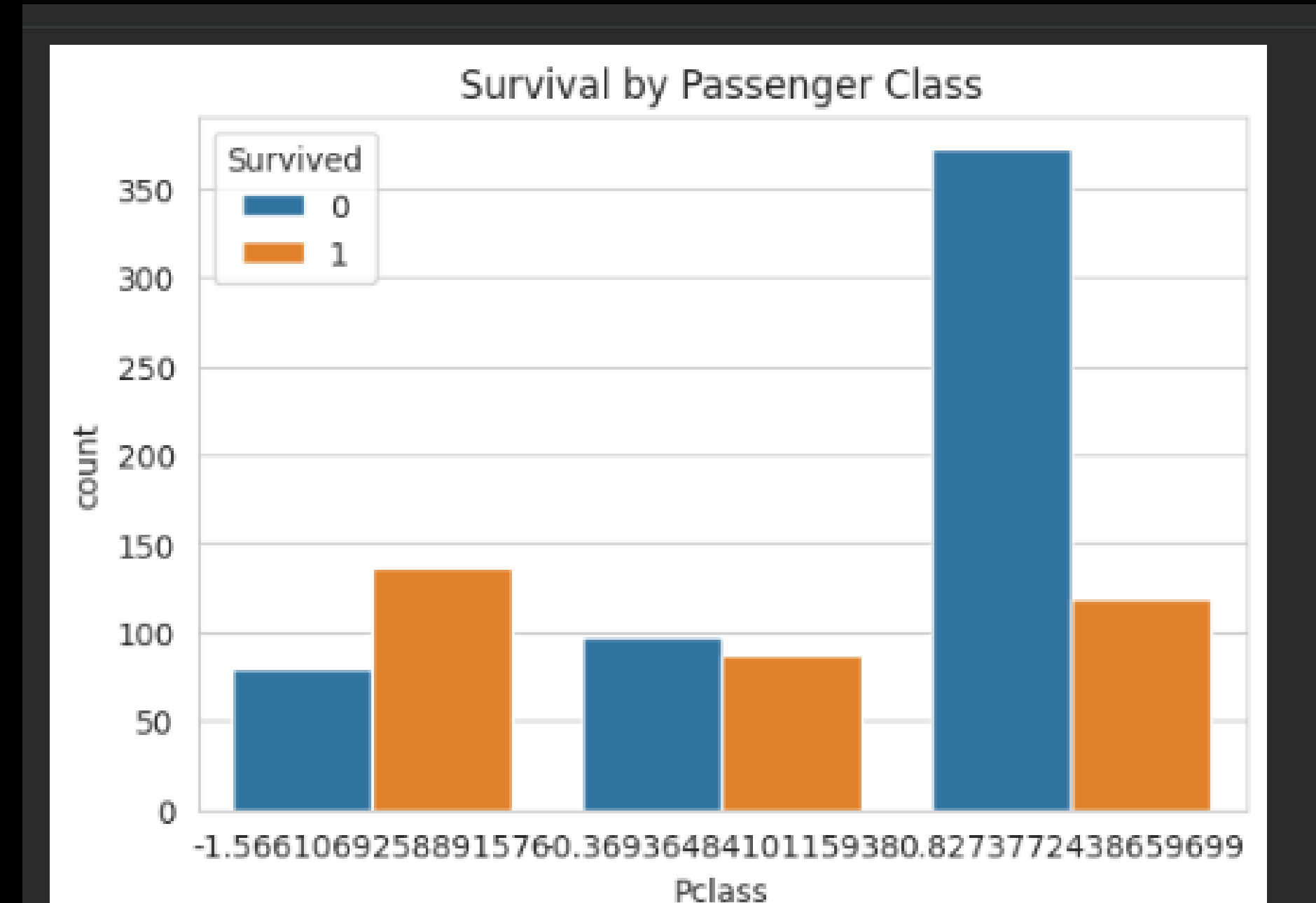
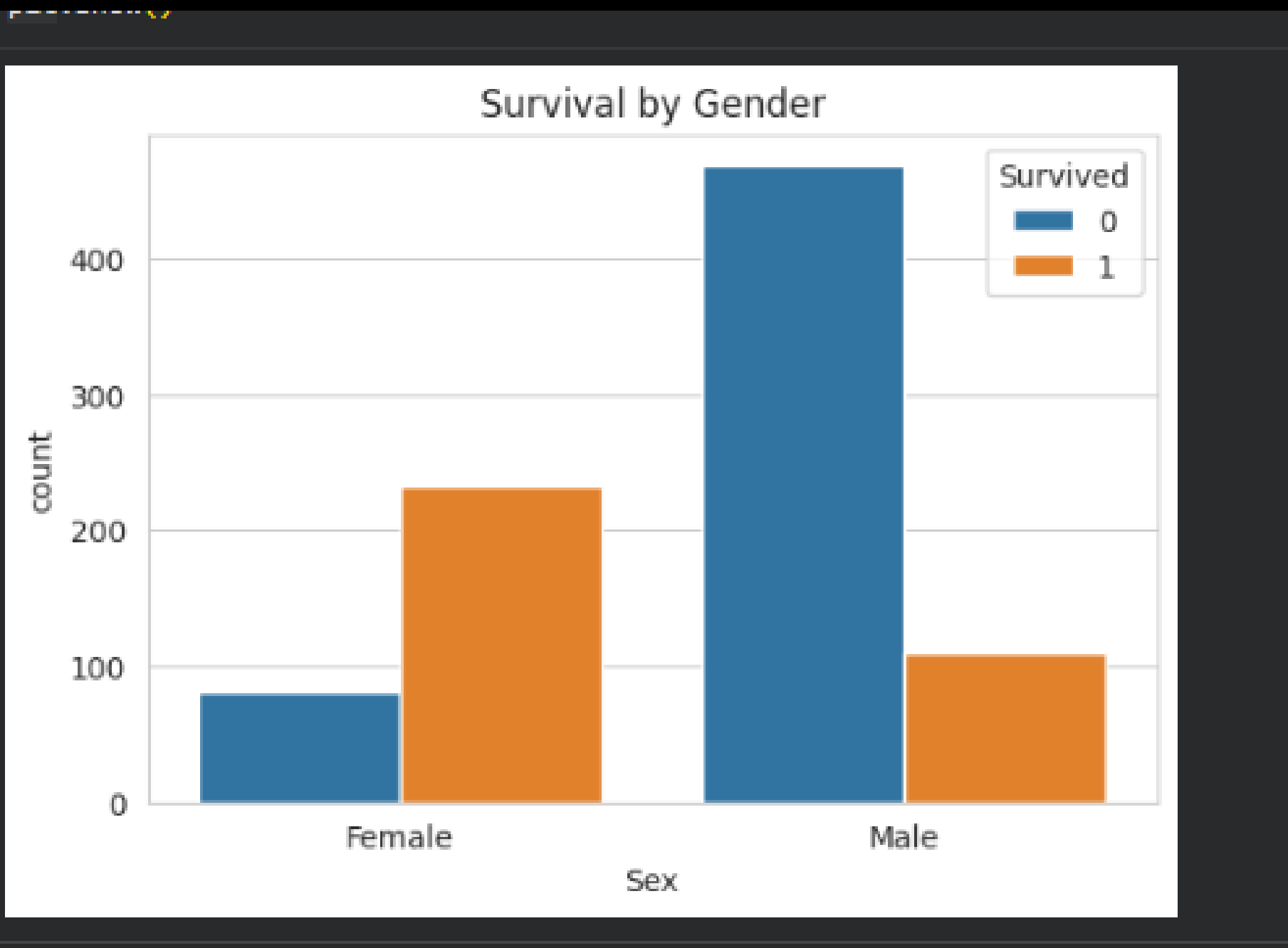
- ✓ Missing Values
- ✓ Duplicate Check
- ✓ Outlier Detection
- ✓ Label Encoding
- ✓ One-Hot Encoding
- ✓ Feature Scaling

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 891 entries, 0 to 890  
Data columns (total 12 columns):  
#   Column      Non-Null Count  Dtype    
--  --        
0   PassengerId  891 non-null    int64    
1   Survived     891 non-null    int64    
2   Pclass       891 non-null    int64    
3   Name         891 non-null    object    
4   Sex          891 non-null    object    
5   Age          714 non-null    float64   
6   SibSp        891 non-null    int64    
7   Parch        891 non-null    int64    
8   Ticket       891 non-null    object    
9   Fare         891 non-null    float64   
10  Cabin        204 non-null    object    
11  Embarked     889 non-null    object    
dtypes: float64(2), int64(5), object(5)  
memory usage: 83.7+ KB
```



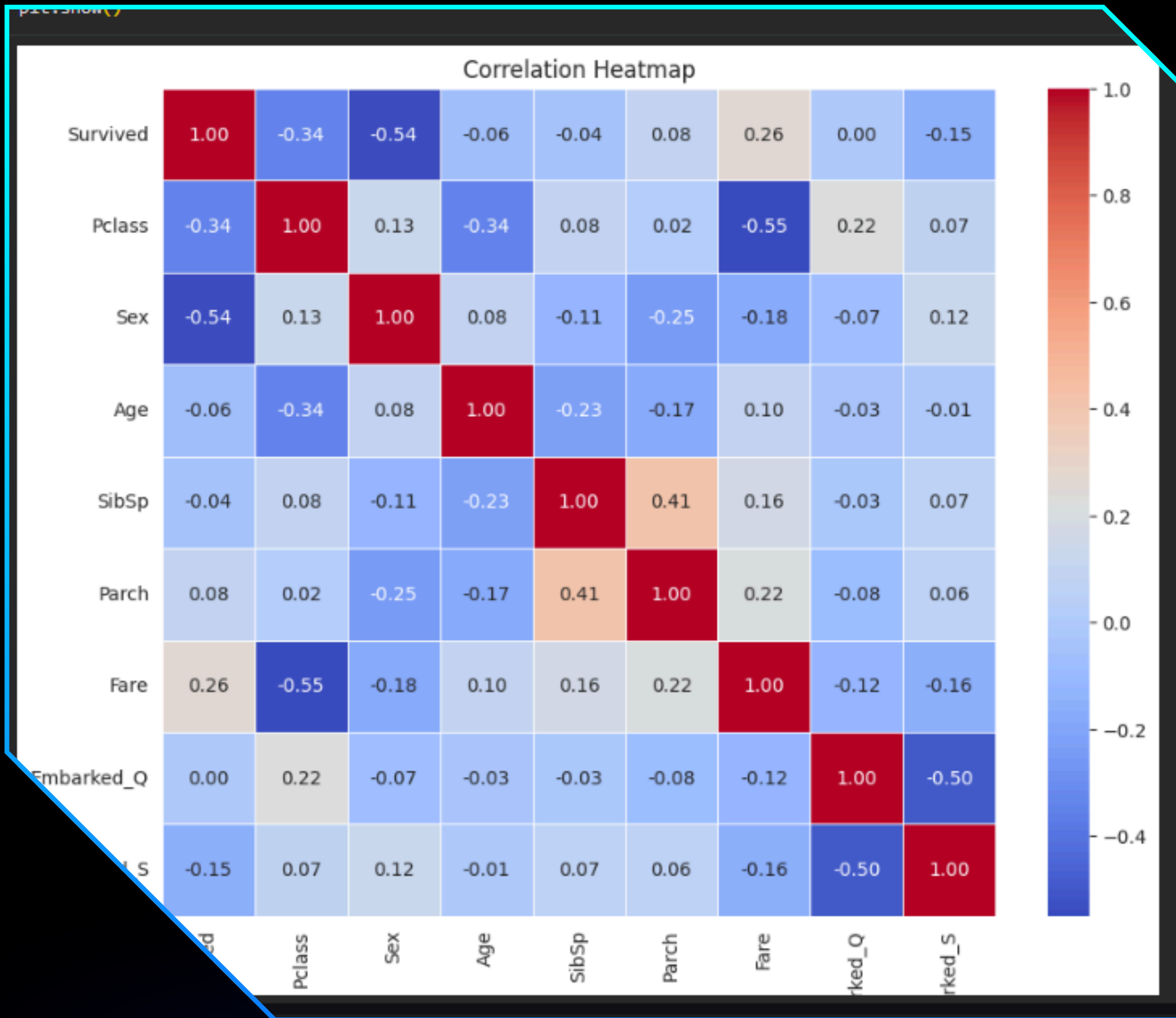
Exploratory Data Analysis



Female passengers had significantly higher survival rates.

Correlation Analysis

Correlation analysis helped identify important relationships between variables for future machine learning models.



Key Insights

Female passengers survived more.

First Class had higher survival.

Fare showed highest variation.

Most passengers travelled alone.

Dataset is now ML Ready.

